

Quartic Analysis

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We're given

- $\mathbf{X}$: an N cells $\times$ P odours matrix of observed input responses,
- $\mathbf{C}$: a P cells $\times$ P cells observed output covariance matrix,
- The regularization parameter $\lambda = \lambda_0 \frac{P^2}{N}$, where λ_0 is the value we sweep in the paper.
- The mean subtraction operation, $\mathbf{J} = \mathbf{I} - N^{-1}\mathbf{1}\mathbf{1}^T$.
- The prediction error $\mathbf{E}(\mathbf{z}) = \mathbf{X}^T[\mathbf{z}]\mathbf{J}[\mathbf{z}]\mathbf{X} - \mathbf{C}$

Our loss function is

$$L(\mathbf{z}) = \frac{1}{2}\|\mathbf{E}(\mathbf{z})\|_F^2 + \frac{\lambda}{2}\|\mathbf{z} - \mathbf{1}\|_2^2, \quad (1) \quad \{\text{eq:loss}\}$$

where $[\cdot]$ converts its argument to a diagonal matrix.

We want to view this from the perspective of a single unit z_i . To do that, we rewrite $\mathbf{E}(\mathbf{z})$ to isolate that unit. Examining its components,

$$\begin{aligned} \mathbf{X}^T[\mathbf{z}]\mathbf{J}[\mathbf{z}]\mathbf{X} &= \mathbf{X}^T[\mathbf{z}^2]\mathbf{X} - N^{-1}\mathbf{X}^T\mathbf{z}\mathbf{z}^T\mathbf{X} \\ \mathbf{X}^T[\mathbf{z}^2]\mathbf{X} &= \sum_{j=1}^N \mathbf{x}_j \mathbf{x}_j^T z_j^2 = \mathbf{x}_i \mathbf{x}_i^T z_i^2 + \sum_{j \neq i} \mathbf{x}_j \mathbf{x}_j^T z_j^2 \\ \mathbf{m} \triangleq \mathbf{X}^T \mathbf{z} &= \sum_j \mathbf{x}_j z_j = \mathbf{x}_i z_i + \sum_{j \neq i} \mathbf{x}_j z_j = \mathbf{x}_i z_i + \mathbf{m}_i \\ \mathbf{E}(\mathbf{z}) &= \mathbf{x}_i \mathbf{x}_i^T z_i^2 + \sum_{j \neq i} \mathbf{x}_j \mathbf{x}_j^T z_j^2 - N^{-1}(\mathbf{x}_i z_i + \mathbf{m}_i)(\mathbf{x}_i z_i + \mathbf{m}_i)^T - \mathbf{C} \\ &= \mathbf{x}_i \mathbf{x}_i^T z_i^2 - N^{-1}(\mathbf{x}_i \mathbf{m}_i^T z_i + \mathbf{m}_i \mathbf{x}_i^T z_i + \mathbf{x}_i \mathbf{x}_i^T z_i^2) + \sum_{j \neq i} \mathbf{x}_j \mathbf{x}_j^T z_j^2 - N^{-1} \mathbf{m}_i \mathbf{m}_i^T - \mathbf{C} \\ &= \alpha \mathbf{x}_i \mathbf{x}_i^T z_i^2 - N^{-1}(\mathbf{x}_i \mathbf{m}_i^T + \mathbf{m}_i \mathbf{x}_i^T) z_i + \mathbf{E}_i. \end{aligned}$$

Computing the goodness-of-fit term of the loss,

$$\begin{aligned} \text{tr}(\mathbf{E}(\mathbf{z})^T \mathbf{E}(\mathbf{z})) &= \alpha^2 \|\mathbf{x}_i\|_2^4 z_i^4 - 4\alpha N^{-1} \|\mathbf{x}_i\|_2^2 \mathbf{m}_i^T \mathbf{x}_i z_i^3 + 2\alpha \mathbf{x}_i^T \mathbf{E}_i \mathbf{x}_i z_i^2 \\ &\quad + 2N^{-2} z_i^2 (\|\mathbf{x}_i\|_2^2 \|\mathbf{m}_i\|_2^2 + (\mathbf{x}_i^T \mathbf{m}_i)^2) - 4N^{-1} z_i \mathbf{x}_i^T \mathbf{E}_i \mathbf{m}_i + \text{tr}(\mathbf{E}_i^T \mathbf{E}_i). \end{aligned}$$

Our loss function in terms of z_i is then

$$\begin{aligned}
L(z_i) = & \frac{1}{2}\alpha^2\|\mathbf{x}_i\|_2^4 z_i^4 \\
& - \frac{2\alpha}{N}\|\mathbf{x}_i\|_2^2 \mathbf{m}_i^T \mathbf{x}_i z_i^3 \\
& + \left(\alpha \mathbf{x}_i^T \mathbf{E}_i \mathbf{x}_i + \frac{\|\mathbf{x}_i\|_2^2 \|\mathbf{m}_i\|_2^2 + (\mathbf{x}_i^T \mathbf{m}_i)^2}{N^2} + \frac{\lambda}{2} \right) z_i^2 \\
& - \frac{2\mathbf{x}_i^T \mathbf{E}_i \mathbf{m}_i + N\lambda}{N} z_i \\
& + \frac{\text{tr}(\mathbf{E}_i^T \mathbf{E}_i) + \lambda(1 + \sum_{i \neq j} (z_j - 1)^2)}{2}.
\end{aligned}$$

The trailing constant doesn't matter for the minimization, so we'll drop it. We'll express the remaining coefficients in terms of a few geometric quantities:

- The raw power of the population mean $m \triangleq \mathbf{m}_i^T \mathbf{m}_i$;
- The raw channel power, $p_i \triangleq \mathbf{x}_i^T \mathbf{x}_i$;
- The residual channel power, $q_i \triangleq \mathbf{x}_i^T \mathbf{E}_i \mathbf{x}_i$;
- The raw alignment with the population mean, $a_i \triangleq \mathbf{x}_i^T \mathbf{m}_i$;
- The residual alignment with the population mean, $b_i \triangleq \mathbf{x}_i^T \mathbf{E}_i \mathbf{m}_i$;
- The curvature at the origin, $\kappa_i \triangleq \alpha q_i + \frac{p_i m + a_i^2}{N^2}$.

In these terms, and dropping the constant,

$$L(z_i) = \frac{1}{2}\alpha^2 p_i^2 z_i^4 - \frac{2\alpha}{N} p_i a_i z_i^3 + \left(\kappa_i + \frac{\lambda}{2} \right) z_i^2 - \left(\frac{2b_i}{N} + \lambda \right) z_i \quad (2) \quad \{\text{eq:Lzi}\}$$

$$\triangleq A z_i^4 + B z_i^3 + C z_i^2 + D z_i. \quad (3) \quad \{\text{eq:Lzi}\}$$

To better understand this quartic, we will change coordinates. First, we'll rescale by A ,

$$\begin{aligned}
L(z) & \propto z^4 + \frac{B}{A} z^3 + \frac{C}{A} z^2 + \frac{D}{A} z \\
& \triangleq z^4 + b z^3 + c z^2 + d z.
\end{aligned}$$

Next, we'll shift to a coordinate $s = z - \bar{z}$ in which the cubic disappears. Expressing the loss as a function $\tilde{L}$ of s , we note that for a monic quartic with no cubic, the third derivative should be $24s$.

$$\tilde{L}(s) = (s + \bar{z})^4 + b(s + \bar{z})^3 + c(s + \bar{z})^2 + d(s + \bar{z}).$$

$$\tilde{L}'''(s) = 24(s + \bar{z}) + 6b = 24s \implies \boxed{\bar{z} = -\frac{b}{4}}.$$

So, in the shifted and scaled coordinates, the loss function is

$$\tilde{L}(s) = \left(s - \frac{b}{4} \right)^4 + c \left(s - \frac{b}{4} \right)^2 + d \left(s - \frac{b}{4} \right)$$

We can expand out the first equation to get the coefficients. But an easier way is to Taylor expand our loss function $L(z)$ to get a polynomial in s :

$$\tilde{L}(s) = L(s + \bar{z}) = L(\bar{z}) + L'(\bar{z})s + \frac{1}{2}L''(\bar{z})s^2 + \frac{1}{24}L'''(\bar{z})s^4.$$

Then, $L''''(\bar{z}) = 24$. The other derivatives are

$$\begin{aligned} L'(\bar{z}) &= 4\bar{z}^3 + 3b\bar{z}^2 + 2c\bar{z} + d = -\frac{b^3}{16} + 3\frac{b^3}{16} - \frac{bc}{2} + d = \frac{b^3}{8} - \frac{bc}{2} + d \\ L''(\bar{z}) &= 12\bar{z}^2 + 6b\bar{z} + 2c = 12\frac{b^2}{16} - 6\frac{b^2}{4} + 2c = -\frac{3}{4}b^2 + 2c. \end{aligned}$$

So we arrive at

$$\begin{aligned} \tilde{L}(s) &= s^4 + \left(c - \frac{3}{8}b^2\right)s^2 + \left(d - \frac{bc}{2} + \frac{b^3}{8}\right)s + \text{const} \\ &\triangleq s^4 + \tilde{c}s^2 + \tilde{d}s + \text{const}. \end{aligned} \tag{4}$$

We can see that $\tilde{L}$ combines a convex quartic with a parabola. If the parabola is also convex, so $\tilde{c} \geq 0$, then $\tilde{L}$ will also be convex, and have only one root. If the parabola is concave (pointing down), so $\tilde{c} < 0$, then $\tilde{L}$ may have one or two roots, depending on the tilt imposed by the linear term.

To determine the location of the root(s), we set the derivative to 0:

$$0 = \tilde{L}'(s) = 4s^3 + 2\tilde{c}s + \tilde{d} = s^3 + \frac{\tilde{c}}{2}s + \frac{\tilde{d}}{4} \triangleq s^3 + gs + h. \tag{5}$$

What do the quantities g , proportional $\tilde{c}$, and $h \propto \tilde{d}$, represent? Since

$$\frac{C}{A} = \frac{2}{\alpha^2 N^2 p_i^2} (N^2 \alpha q_i + p_i m + a_i^2), \quad \frac{B}{A} = -\frac{4a_i}{\alpha N p_i} \implies \left(\frac{B}{A}\right)^2 = \frac{16a_i^2}{\alpha^2 N^2 p_i^2},$$

we get

$$\tilde{c} = \frac{C}{A} - \frac{3}{8} \left(\frac{B}{A}\right)^2 = \boxed{2 \frac{N^2 \alpha q_i + p_i m - 2a_i^2}{\alpha^2 N^2 p_i^2}}.$$

To interpret this term, we consider its numerator,

$$\begin{aligned} N^2 \alpha q_i + p_i m - 2a_i^2 &= N^2 \alpha \mathbf{x}_i^T \mathbf{E}_i \mathbf{x}_i + \|\mathbf{x}_i\|_2^2 \|\mathbf{m}_i\|_2^2 - 2(\mathbf{x}_i^T \mathbf{m}_i)^2 \\ &= \|\mathbf{x}_i\|_2^2 \left(N^2 \alpha \frac{\mathbf{x}_i^T \mathbf{E}_i \mathbf{x}_i}{\|\mathbf{x}_i\|_2^2} + \|\mathbf{m}_i\|_2^2 - 2 \frac{(\mathbf{x}_i^T \mathbf{m}_i)^2}{\|\mathbf{x}_i\|_2^2} \right) \\ &= \|\mathbf{x}_i\|_2^2 (N^2 \alpha \hat{\mathbf{x}}_i^T \mathbf{E}_i \hat{\mathbf{x}}_i + \|\mathbf{m}_i\|_2^2 - 2(\hat{\mathbf{x}}_i^T \mathbf{m}_i)^2) \\ &= N^2 \alpha \|\mathbf{x}_i\|_2^2 \left(\hat{\mathbf{x}}_i^T \mathbf{E}_i \hat{\mathbf{x}}_i + \frac{\|\mathbf{m}_i^\perp\|_2^2 - \|\mathbf{m}_i^\parallel\|_2^2}{N^2 \alpha} \right) \\ &= N^2 \alpha \|\mathbf{x}_i\|_2^2 \left(\hat{\mathbf{x}}_i^T \mathbf{E}_i \hat{\mathbf{x}}_i + \frac{\|\mathbf{m}_i\|_2^2 (1 - 2\cos(\theta_i)^2)}{N^2 \alpha} \right) \\ &= N^2 \alpha \|\mathbf{x}_i\|_2^2 \left(\hat{\mathbf{x}}_i^T \mathbf{E}_i \hat{\mathbf{x}}_i - \frac{\|\mathbf{m}_i\|_2^2 \cos(2\theta_i)}{N^2 \alpha} \right). \end{aligned}$$

where we've defined

$$\mathbf{m}_i^\parallel \triangleq \mathbf{m}_i^T \hat{\mathbf{x}}_i \hat{\mathbf{x}}_i, \quad \mathbf{m}_i^\perp \triangleq \mathbf{m}_i - \mathbf{m}_i^\parallel$$

as the components of $\mathbf{m}_i$ parallel and perpendicular to $\mathbf{x}_i$. Therefore,

$$\|\mathbf{x}_i\|^2 g = \|\mathbf{x}_i\|^2 \frac{\tilde{c}}{2} = \alpha^{-1} \hat{\mathbf{x}}_i^T \mathbf{E}_i \hat{\mathbf{x}}_i - \frac{\|\mathbf{m}_i\|_2^2 \cos(2\theta_i)}{\alpha^2 N^2}.$$

Defining

$$\tilde{\mathbf{E}}_i \triangleq \frac{1}{\alpha} \mathbf{E}_i, \quad \boldsymbol{\mu}_i \triangleq \frac{\mathbf{m}_i}{N-1},$$

this simplifies to

$$\|\mathbf{x}_i\|_2^2 g = \hat{\mathbf{x}}_i^T \tilde{\mathbf{E}}_i \hat{\mathbf{x}}_i - \|\boldsymbol{\mu}_i\|_2^2 \cos(2\theta_i).$$

Next, since

$$\frac{D}{A} = -\frac{4b_i}{N\alpha^2 p_i^2} = -\frac{4Nb_i}{N^2\alpha^2 p_i^2}, \quad \frac{BC}{2A^2} = -\frac{4\alpha p_i a_i \kappa_i}{N\alpha^4 p_i^4} = \frac{-4a_i \kappa_i}{N\alpha^3 p_i^3}, \quad \frac{1}{8} \left(\frac{B}{A}\right)^3 = -\frac{8a_i^3}{\alpha^3 N^3 p_i^3},$$

we get

$$\begin{aligned} \tilde{d} &= d - \frac{bc}{2} + \frac{b^3}{8} = -\frac{4b_i}{N\alpha^2 p_i^2} + \frac{4a_i \kappa_i}{N\alpha^3 p_i^3} - \frac{8a_i^3}{\alpha^3 N^3 p_i^3} \\ &= 4 \frac{N^2(a_i \kappa_i - \alpha b_i p_i) - 2a_i^3}{\alpha^3 N^3 p_i^3} = 4 \frac{N^2(a_i(\alpha q_i + \frac{p_i m + a_i^2}{N^2}) - \alpha b_i p_i) - 2a_i^3}{\alpha^3 N^3 p_i^3} \\ &= \boxed{4 \frac{N^2\alpha(q_i a_i - b_i p_i) + a_i p_i m - a_i^3}{\alpha^3 N^3 p_i^3}}. \end{aligned}$$

To interpret this quantity we again consider it's numerator,

$$\begin{aligned} N^2\alpha(q_i a_i - b_i p_i) + a_i p_i m - a_i^3 &= N^2\alpha \left((\mathbf{x}_i^T \mathbf{E}_i \mathbf{x}_i)(\mathbf{x}_i^T \mathbf{m}_i) - (\mathbf{x}_i^T \mathbf{E}_i \mathbf{m}_i)(\mathbf{x}_i^T \mathbf{x}_i) + \frac{(\mathbf{x}_i^T \mathbf{m}_i)(\mathbf{x}_i^T \mathbf{x}_i)(\mathbf{m}_i^T \mathbf{m}_i) - (\mathbf{x}_i^T \mathbf{m}_i)^3}{N^2\alpha} \right) \\ &= N^2\alpha \|\mathbf{x}_i\|^3 \|\mathbf{m}_i\| \left((\hat{\mathbf{x}}_i^T \mathbf{E}_i \hat{\mathbf{x}}_i)(\hat{\mathbf{x}}_i^T \hat{\mathbf{m}}_i) - \hat{\mathbf{x}}_i^T \mathbf{E}_i \hat{\mathbf{m}}_i + \frac{\hat{\mathbf{x}}_i^T \hat{\mathbf{m}}_i \|\mathbf{m}_i\|^2 - (\hat{\mathbf{x}}_i^T \hat{\mathbf{m}}_i)^3 \|\mathbf{m}_i\|^2}{N^2\alpha} \right) \\ &= N^2\alpha \|\mathbf{x}_i\|^3 \|\mathbf{m}_i\| \left(\hat{\mathbf{x}}_i^T \mathbf{E}_i \hat{\mathbf{x}}_i \cos(\theta_i) - \hat{\mathbf{x}}_i^T \mathbf{E}_i \hat{\mathbf{m}}_i + \cos(\theta_i) \frac{\|\mathbf{m}_i\|^2 - \cos(\theta_i)^2 \|\mathbf{m}_i\|^2}{N^2\alpha} \right) \\ &= -N^2\alpha \|\mathbf{x}_i\|^3 \|\mathbf{m}_i\| \left(\hat{\mathbf{x}}_i^T \mathbf{E}_i \hat{\mathbf{m}}_i^\perp \sin(\theta_i) - \frac{\|\mathbf{m}_i\|^2 \sin(\theta_i)^2 \cos(\theta_i)}{N^2\alpha} \right) \\ &= -N^2\alpha \|\mathbf{x}_i\|^3 \|\mathbf{m}_i^\perp\| \left(\hat{\mathbf{x}}_i^T \mathbf{E}_i \hat{\mathbf{m}}_i^\perp - \frac{\|\mathbf{m}_i\|^2 \sin(2\theta_i)}{2N^2\alpha} \right) \end{aligned}$$

where $\hat{\mathbf{m}}_i^\perp$ is the unit vector along the component of $\mathbf{m}_i$ perpendicular to $\mathbf{x}_i$. Therefore

$$\begin{aligned} \|\mathbf{x}_i\|^3 h &= \|\mathbf{x}_i\|^3 \frac{\tilde{d}}{4} = -\frac{1}{N\alpha^2} \|\mathbf{m}_i^\perp\| \left(\hat{\mathbf{x}}_i^T \mathbf{E}_i \hat{\mathbf{m}}_i^\perp - \frac{\|\mathbf{m}_i\|^2 \sin(2\theta_i)}{2N^2\alpha} \right) \\ &= -\frac{1}{N\alpha} \|\mathbf{m}_i^\perp\| \left(\hat{\mathbf{x}}_i^T \tilde{\mathbf{E}}_i \hat{\mathbf{m}}_i^\perp - \frac{\|\mathbf{m}_i\|^2 \sin(2\theta_i)}{2N^2\alpha^2} \right), \end{aligned}$$

so we arrive at

$$\boxed{\|\mathbf{x}_i\|^3 h = -\|\boldsymbol{\mu}_i^\perp\| \left(\hat{\mathbf{x}}_i^T \tilde{\mathbf{E}}_i \hat{\boldsymbol{\mu}}_i^\perp - \frac{1}{2} \|\boldsymbol{\mu}_i\|^2 \sin(2\theta_i) \right)}.$$

We can also express this entirely in terms of $\boldsymbol{\mu}_i^\perp$ as

$$\|\mathbf{x}_i\|^3 h = -\|\boldsymbol{\mu}_i^\perp\| \left(\hat{\mathbf{x}}_i^T \tilde{\mathbf{E}}_i \hat{\boldsymbol{\mu}}_i^\perp - \|\boldsymbol{\mu}_i^\perp\|^2 \cot(\theta_i) \right).$$

To solve for the roots of equation 5, we use the triple angle formula:

$$4\cos^3(\theta) - 3\cos(\theta) - \cos(3\theta) = 0.$$

Let $s = K\cos(\theta)$. Then our derivative equation is

$$K^3 \cos(\theta)^3 + Kg \cos(\theta) + h = 0.$$

Dividing out by $K^3/4$, we get

$$4 \cos(\theta)^3 + \frac{4g}{K^2} \cos(\theta) + \frac{4h}{K^3} = 0.$$

Now if $g \leq 0$, we can set K such that

$$\frac{4g}{K^2} = -3 \implies K = 2\sqrt{|g|/3}.$$

Then we get

$$\cos(3\theta) = -\frac{4h}{K^3} = -\frac{4h}{K^2} \frac{1}{K} = -\frac{3h}{2|g|} \sqrt{\frac{3}{|g|}} = \frac{3h}{2g} \sqrt{\frac{3}{-g}} \triangleq u.$$

This equation will have solution if $|u| \leq 1$, equivalently

$$1 \geq u^2 = \frac{27h^2}{-4g^3} \iff 27h^2 \leq 4(-g)^3,$$

with equality determining the *cusp*. The solutions will satisfy

$$3\theta = \arccos(u) + 2\pi k \implies \theta = \frac{1}{3} \arccos(u) + \frac{2\pi k}{3}, \quad k \in \{0, 1, 2\},$$

corresponding to the two wells and one ridge.

Which k corresponds to the ridge, and which to the valleys? When $u = 0$, there is no tilt in the quartic and the wells will be symmetric around 0. In this case,

$$\theta = \frac{1}{3} \arccos(0) + \frac{2\pi k}{3} = \frac{\pi}{6} + \frac{2\pi k}{3} = \left\{ \frac{\pi}{6}, \frac{5\pi}{6}, \frac{3\pi}{2} \right\}.$$

Then

$$\cos(\theta) = \{\sqrt{3}/2, -\sqrt{3}/2, 0\}.$$

So, assuming this ordering holds at non-zero u , $k = 0$ gives the solution when the tilt is negative, $k = 1$ gives the solution for positive tilt, and $k = 2$ gives the ridge.

Now we're interested in how large the cosine term can get. As the tilt becomes more negative, the minimum at $\sqrt{3}/2$ will move to the right. How far forward will it go? Negative tilt corresponds to negative h and hence positive u since g is negative in this regime. Hence the maximally negative tilt, occurring when the ridge is about disappear, will correspond to the $u = 1$ case. At that point

$$\theta = \frac{1}{3} \arccos(1) + \frac{2\pi k}{3} = \left\{ 0, \frac{2\pi}{3}, \frac{4\pi}{3} \right\}$$

So then

$$\cos(\theta) = \{1, -1/2, -1/2\},$$

corresponding to a global minimum at 1 and an inflection point at $-1/2$.

So, when $|u| \leq 1$, we can represent the solution as

$$s^* = \frac{\sqrt{3}}{2} K \text{sign}(u) f(|u|), \quad f \in \left[1, \frac{2}{\sqrt{3}} \right].$$

Plugging in our expression for K simplifies this to

$$s^* = \sqrt{|g|} \text{sign}(u) f(|u|).$$

Now let's consider the case when $g > 0$. In that case, the relevant triple-angle formula is for $\sinh$,

$$4 \sinh^3(\theta) + 3 \sinh(\theta) - \sinh(3\theta) = 0.$$

If we set $s = K \sinh(\theta)$, our depressed cubic becomes

$$K^3 \sinh(\theta)^3 + Kg \sinh(\theta) + h = 0.$$

Dividing by $K^3/4$ this gives

$$4 \sinh(\theta)^3 + \frac{4g}{K^2} \sinh(\theta) + \frac{4h}{K^3} = 0.$$

Setting

$$\frac{4g}{K^2} = 3 \implies K = 2\sqrt{g/3}$$

gives

$$\sinh(3\theta) = -\frac{4h}{K^3} = -\frac{4h}{K^2} \frac{1}{K} = -3 \frac{4h}{4g} \frac{1}{K} = -\frac{3h}{2g} \sqrt{\frac{3}{g}} \triangleq v.$$

We then get $\theta = \frac{1}{3} \operatorname{arcsinh}(v)$, so

$$s^* = K \sinh(\theta) = 2\sqrt{\frac{g}{3}} \sinh\left(\frac{1}{3} \operatorname{arcsinh}(v)\right).$$

When v is small,

$$s^* \approx \frac{1}{3} K v = -\frac{h}{g}.$$

When v is large,

$$s^* \approx -h^{1/3}.$$

Complexifying

Rather than treating each of these cases separately, we can view them all in a unified manner by consider the roots of our depressed cubic in the complex plane.

In that setting, the triple angle formula is

$$w^3 + w^{-3} = (w + w^{-1})^3 - 3(w + w^{-1}).$$

We now parameterize

$$s = \frac{K}{2}(w + w^{-1}).$$

Our cubic equation then becomes

$$\frac{K^3}{8}(w + w^{-1})^3 + \frac{Kg}{2}(w + w^{-1}) + h = 0.$$

Normalizing,

$$(w + w^{-1})^3 + \frac{4g}{K^2}(w + w^{-1}) + \frac{8h}{K^3} = 0.$$

Setting K so that

$$\frac{4g}{K^2} = -3 \implies K = \sqrt{\frac{-4g}{3}} = 2\sqrt{\frac{-g}{3}},$$

Making that substitution gives

$$w^3 + w^{-3} = -\frac{8h}{K^3} \implies w^3 + w^{-3} + \frac{8h}{K^3} = 0.$$

Then letting $W = w^3$, we have

$$W + \frac{1}{W} + \frac{8h}{K^3} = 0.$$

Multiplying by W , we get

$$W^2 + 2\frac{4h}{K^3}W + 1 = 0.$$

Since the constant coefficient is 1, we have that that two roots of this equation are inverses of each other.

Defining

$$u \triangleq -\frac{4h}{K^3} = -\frac{4h}{K^2} \frac{1}{K} = \frac{3h}{2g} \sqrt{\frac{3}{-g}}, \quad u^2 = -\frac{27h^2}{4g^3},$$

we get

$$W = \frac{2u \pm \sqrt{4u^2 - 4}}{2} = u \pm \sqrt{u^2 - 1}.$$

Now each value of W will gives us three values of w . There are two values of W , so this will yield six total values of w . However, these can be put in three pairs, where each element of the pair is the inverse of the other. And s is invariant to inverses, so $s(w) = s(w^{-1})$, and we get three values of s as the roots of the cubic, as expected.

Unified derivation

To find the roots s , take wolog $W = u + \sqrt{u^2 - 1}$. Then

$$w = \sqrt[3]{W} = \sqrt[3]{W} e^{j2\pi} = e^{\ln(W) + j2\pi k/3}, \quad k \in \{0, 1, 2\}.$$

Then

$$s_k = K \frac{w + w^{-1}}{2} = K \cosh \left(\frac{1}{3} \ln(W) + j \frac{2\pi k}{3} \right).$$

To determine the $\ln$ term, we have

$$\ln(u + \sqrt{u^2 - 1}) = x \implies e^x = u + \sqrt{u^2 - 1}, \quad e^{-x} = u - \sqrt{u^2 - 1},$$

the latter because the two roots of the quadratic multiply to 1. Then

$$e^x + e^{-x} = 2u \implies u = \cosh(x) \implies x = \operatorname{arccosh}(u),$$

so we arrive at the unified expression

$$s_k = K \cosh \left(\frac{1}{3} \operatorname{arccosh}(u) + j \frac{2\pi k}{3} \right).$$

Now recall that

$$K \propto \sqrt{-g}, \quad u = -4h/K^3.$$

If $g > 0$ then K is imaginary, which means u is imaginary, say $j\nu$ for real ν . Let $\operatorname{arccosh}(j\nu) = r + j\phi$. We need $e^r e^{j\phi} + e^{-r} e^{-j\phi} = j2\nu$. Letting $\phi = \pi/2$, we get $j(e^r - e^{-r}) = j2\nu \implies j \sinh(r) = j\nu \implies r = \operatorname{arcsinh}(\nu)$, so

$$s_k = K \cosh \left(\frac{1}{3} \operatorname{arcsinh}(\operatorname{Im}(u)) + j \frac{\pi}{6} + j \frac{2\pi k}{3} \right).$$

If $g < 0$, the K is real. If $|u| > 1$, the $\operatorname{arccosh}$ formula from the unified express goes through as is.

If $|u| \leq 1$: then we need x such that $\cosh(x) = \cos(jx) = u \implies jx = \arccos(u) \implies x = j \arccos(u)$, so we get

$$s_k = K \cosh \left(j \frac{\arccos(u) + 2\pi k}{3} \right) = K \cos \left(\frac{\arccos(u) + 2\pi k}{3} \right).$$

Approximations

$g < 0, u^2 \leq 1$

- Here K is real, so u is real.
- Since $|u| \leq 1$, $\operatorname{arccosh}(u) = j \operatorname{arccos}(u)$.
- Then $s_k = K \cosh(j(\operatorname{arccos}(u) + 2\pi k)/3) = K \cos \frac{\operatorname{arccos}(u) + 2\pi k}{3}$.
- Three real roots, one will be the saddle.
- At $u = 0$, $\operatorname{arccos}(u) = \pi/2$, $\theta_k = \pi/6, 5\pi/6, 9\pi/6 = 3\pi/2 = -\pi/2$, $\cos(\theta_k) = \{\sqrt{3}/2, -\sqrt{3}/2, 0\}$.
- For positive u , the tilt h is negative, so the minimum will be positive, starting near $\sqrt{3}/2$.
- at $u = 1$, $\operatorname{arccos}(u) = 0$, $\theta_k = \{0, 2\pi/3, -2\pi/3\}$, so $\cos(\theta_k) = \{1, 1/2, 1/2\}$. Notice that the shoulder and the higher well have now merged.
- So in this regime, the minimum will be at

$$s^* = K f(|u|) \operatorname{sgn}(u), \quad f(|u|) \in [\sqrt{3}/2, 1] \implies s^* \approx K \frac{\sqrt{3}}{2} \operatorname{sgn}(u) = \operatorname{sgn}(u) \sqrt{-g}.$$

$g < 0, u^2 \geq 1$

- Here K is real, so u is real.
- $|u| \geq 1$ so $\operatorname{arccosh}(u) = \ln(u \pm \sqrt{u^2 - 1})$.
- This gives a solution greater than one, and its inverse.
- We take the one that's greater than one, as that's the trajectory of the minimum.
- Now if $u = 1 + \varepsilon$, then $\ln$'s argument is approximately $1 + \sqrt{2\varepsilon}$, so $\operatorname{arccosh}(u) = \ln(\dots) \approx \sqrt{2\varepsilon}$, and

$$s^* \approx K \cosh(\sqrt{2\varepsilon}/3) \operatorname{sgn}(u) \approx \frac{2}{\sqrt{3}} \operatorname{sgn}(u) \sqrt{-g}.$$

- If $u^2 \gg 1$, then $\ln(u + \sqrt{u^2 - 1}) \approx \ln(2u)$, so

$$s^* \approx K \operatorname{sgn}(u) \cosh(\ln(2u)/3) \approx \frac{K \operatorname{sgn}(u)}{2} \sqrt[3]{2u} = \operatorname{sgn}(u) \frac{K}{2} \sqrt[3]{2} \frac{\sqrt[3]{4} \sqrt[3]{|h|}}{K} = \operatorname{sgn}(u) \sqrt[3]{|h|}.$$

Case by case derivation

Let's consider the various cases.

Case 1: $g < 0$

First, let's assume $g < 0$, so K is real. In that case, $u^2 \geq 0$.

If $|u| \leq 1$, then the square-root is pure imaginary, so $|W| = 1$ and W is on the unit circle. So

$$W = e^{j(3\theta + 2\pi k)} \implies w = e^{j\theta_k}, \quad \theta_k \triangleq \theta + \frac{2\pi k}{3}, \quad k \in \{0, 1, 2\}.$$

This in turn means three real roots

$$s_k = K \cos \left(\theta + \frac{2\pi k}{3} \right), \quad k \in \{0, 1, 2\}, \quad \theta = \frac{1}{3} \operatorname{arccos}(u).$$

If $|u| > 1$, then the square-root in the definition of W is real, so

$$w = W^{1/3} \implies w_k = \sqrt[3]{W} e^{j2\pi k/3}.$$

Then

$$\begin{aligned} s_0 &= K \operatorname{sgn}(W) \frac{|W|^{1/3} + |W|^{-1/3}}{2} = K \operatorname{sgn}(W) \frac{e^{\frac{\ln(|W|)}{3}} + e^{-\frac{\ln(|W|)}{3}}}{2} \\ &= K \operatorname{sgn}(W) \cosh\left(\frac{\ln(|W|)}{3}\right) = K \operatorname{sgn}(W) \cosh\left(\frac{\operatorname{arccosh}|u|}{3}\right). \end{aligned}$$

The next root is

$$s_1 = K \operatorname{sgn}(W) \frac{|W|^{1/3} E[j2\pi/3] + |W|^{-1/3} E[-j2\pi/3]}{2}, \quad E[x] \triangleq e^{jx},$$

while

$$s_2 = K \operatorname{sgn}(W) \frac{|W|^{1/3} E[-j2\pi/3] + |W|^{-1/3} E[j2\pi/3]}{2} = \overline{s_1}.$$

Since the roots of a depressed cubic sum to zero, we get

$$s_{1,2} = -\frac{s_0}{2} \pm j\eta.$$

The imaginary component η should be easy to compute. Since

$$\operatorname{Im}(E[j2\pi/3]) = \frac{\sqrt{3}}{2} = -\operatorname{Im}(E[-j2\pi/3]),$$

$$\eta = \operatorname{Im}(s_1) = K \operatorname{sgn}(W) \frac{\sqrt{3}}{2} \frac{|W|^{1/3} - |W|^{-1/3}}{2} = \frac{\sqrt{3}}{2} K \operatorname{sgn}(W) \sinh\left(\frac{\operatorname{arccosh}|u|}{3}\right).$$

We therefore have, using that $\operatorname{sgn}(W) = \operatorname{sgn}(u)$,

$$s_0 = K \operatorname{sgn}(u) \cosh(\phi), \quad s_{1,2} = \frac{K}{2} \operatorname{sgn}(u) (-\cosh(\phi) \pm j\sqrt{3} \sinh(\phi)), \quad \phi = \frac{1}{3} \operatorname{arccosh}(|u|).$$

Case 2: $g > 0$

Next, let's assume $g > 0$, so K is imaginary. Then $u^2 < 0$, so u is imaginary, and W is pure imaginary, say ja . Since the product of it and its paired root jb is 1,

$$1 = jaja = -1ab \implies b = -1/a,$$

so the roots are $W \in \{ja, -j/a\}$. Each such pair will give the same s . So we can work with one element, and absorb signs into it so that $a > 0$. To ensure this condition holds, we have to take the right root of the quadratic defining W . That root ends up being the positive one: letting $u = j\nu$ for some real ν , we get that

$$u = j\nu, \quad W = ja, \quad a > 0 \implies a = \nu + \sqrt{\nu^2 + 1}.$$

Then since $W = ja = e^{\ln(a) + j(\pi/2 + 2\pi k)}$, we get

$$w = W^{1/3} = e^{\frac{1}{3}\ln(a) + j\theta_k}, \quad \theta_k = \frac{\pi}{6} + \frac{2\pi k}{3}, \quad k \in \{0, 1, 2\}.$$

When $k = 0$, $\theta_0 = \frac{\pi}{6}$, so,

$$j(w_0 + w_0^{-1}) = \sqrt[3]{a} E[\pi/6 + \pi/2] + \sqrt[3]{a^{-1}} E[-\pi/6 + \pi/2] = \sqrt[3]{a} E[2\pi/3] + \sqrt[3]{a^{-1}} E[\pi/3].$$

When $k = 1$, $\theta_1 = \frac{\pi}{6} + \frac{4\pi}{6} = \frac{5\pi}{6}$ so

$$j(w_1 + w_1^{-1}) = \sqrt[3]{a}E[5\pi/6 + \pi/2] + \sqrt[3]{a^{-1}}E[-5\pi/6 + \pi/2] = \sqrt[3]{a}E[-2\pi/3] + \sqrt[3]{a^{-1}}E[-\pi/3].$$

We therefore see that the corresponding roots are complex conjugates of each-other:

$$s_0 = \text{Im}(K) \frac{\sqrt[3]{a}E[2\pi/3] + \sqrt[3]{a^{-1}}E[\pi/3]}{2}, \quad s_1 = \overline{s_0}.$$

For the remaining case of $k = 2$, $\theta_2 = \frac{9\pi}{6} = 3\pi/2 = -\pi/2$, so

$$j(w_2 + w_2^{-1}) = \sqrt[3]{a}[-\pi/2 + \pi/2] + \sqrt[3]{a^{-1}}[\pi/2 + \pi/2] = \sqrt[3]{a} - \sqrt[3]{a^{-1}},$$

so

$$\frac{j(w_2 + w_2^{-1})}{2} = \frac{e^{\ln(a)/3} - e^{-\ln(a)/3}}{2} = \sinh(\ln(a)/3).$$

Since $\ln(\nu + \sqrt{\nu^2 + 1}) = \text{arcsinh}(\nu)$, our expression above simplifies. Combining it with K , we get

$$s_2 = \text{Im}(K) \sinh\left(\frac{1}{3} \text{arcsinh}(\nu)\right).$$

Since the roots of a depressed cubic sum to zero, we get

$$s_{0,1} = -\frac{s_2}{2} \pm j\eta.$$

The imaginary component is easy to compute, since

$$\text{Im}(E[2\pi/3]) = \text{Im}(E[\pi/3]) = \frac{\sqrt{3}}{2}.$$

Then

$$\eta = \frac{\text{Im}(K)}{2} \sqrt{3} \cosh\left(\frac{1}{3} \text{arcsinh}(\nu)\right),$$

so that

$$s_2 = \text{Im}(K) \sinh(\phi), \quad s_{0,1} = \frac{\text{Im}(K)}{2} (-\sinh(\phi) \pm j\sqrt{3} \cosh(\phi)), \quad \phi \triangleq \frac{1}{3} \text{arcsinh}(\text{Im}(u)).$$